

Erdős Problem #854

EP-854

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let n_k denote the k th primorial, i.e. the product of the first k primes.

If $1 = a_1 < a_2 < \cdots a_{\phi(n_k)} = n_k - 1$ is the sequence of integers coprime to n_k , then estimate the smallest even integer not of the form $a_{i+1} - a_i$. Are there $\gg \max_i(a_{i+1} - a_i)$ many even integers of the form $a_{j+1} - a_j$?